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Financial News Sentiment for Stock Price Prediction

Assessing Returns and Managing Exposure in FTSE 100 Equities

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This dissertation is submitted as part requirement for the MSc degree at UCL. It is substantially the result of my own work except where explicitly indicated in the text.

Abstract

This dissertation examines whether financial-news sentiment can help assess or predict equity returns and inform long-only trading decisions. English-language Refinitiv headlines are linked to a fixed sample of 100 FTSE 100 constituents from January to June 2026. FinBERT scores each cleaned headline, and the mean positive-minus-negative probability within a strict information window forms the company signal. Whole-period mean sentiment has a Pearson correlation of 0.436 with constituent return, whereas constituent-normalised headline count is only weakly related to absolute daily movement (Pearson 0.123; Spearman 0.047). The best 24-hour 16:15 threshold rule returns 5.61% at midpoint and 5.31% after paying the quoted spread, below the 5.70% FTSE 100 price return but above the 4.97% equally weighted constituent mean. Daily Top 1 and Top 2 sentiment portfolios return 39.84% and 19.88%, but these results are highly concentrated, sensitive to decision time and exclude additional trading costs. Sentiment is therefore useful as a descriptive measure and as a candidate input to exposure control and stock selection, but high in-sample returns do not establish a general forecasting rule; testing on independent dates against price-only controls, with stamp duty, dividends and point-in-time index membership, is required before any practical claim.

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Chapter 1

Introduction

1.1 Motivation

Financial decisions depend on information that is often first communicated in language: earnings releases, analyst reports, contract announcements and regulatory decisions describe events before investors reduce them to numbers. Sentiment analysis measures whether text conveys a favourable, neutral or adverse financial implication. News can therefore describe company conditions, anticipate a subsequent price adjustment, or signal deterioration that justifies reducing exposure.

1.2 Thesis Statement and Objectives

The thesis statement is that financial-news sentiment may be useful for assessing company performance and controlling exposure even when it does not reliably maximise trading returns. The research question is: to what extent can financial-news sentiment help assess or predict equity returns, and under which trading conditions does that information improve outcomes? Assessment means a same-period association between tone and return. Prediction requires the return to follow the available signal. Investment usefulness additionally requires an explicit rule whose return or downside outcome improves against a stated comparator after the assumed execution costs.

The first objective is to determine whether average news sentiment and coverage intensity are associated with returns, separating whole-period assessment from shorter-horizon prediction. The second is to evaluate whether transparent sentiment rules improve long-only investment outcomes relative to passive ownership. The third is to identify how execution prices, decision time and capital allocation change those outcomes. The fourth is to establish whether any benefit is better characterised as return enhancement, loss avoidance or a combination of the two.

FinBERT is an instrument for producing reproducible textual measurements. A la-

belled comparison assesses its suitability, but success on sentences related to its fine-tuning data is not an independent test of generalisation. The central empirical target is the relationship between the measurements and investment outcomes; Appendix D illustrates why a correctly labelled headline need not forecast a return.

1.3 Contributions

First, the study documents the relationship between headline quantity, mean financial tone and returns in a fixed UK large-cap sample. Second, it links an interpretable signal to long-only decisions while making window boundaries, compounding and quoted spreads explicit. Third, it distinguishes stock-level loss avoidance from concentrated portfolio gains and examines whether sentiment merits further testing as an exposure-control input. The positive whole-period association is an extension of established sentiment research to this corpus, market and period, not a claim of first discovery.

These contributions are exploratory. The study compares many parameter values on the same six months, so the highest observed return is the largest of many tested configurations, not an estimate of an invariant optimum, and later chronological testing is needed to establish whether the relationships persist.

1.4 Methodology Overview

The experiments add decision rules in layers: coverage and mean tone are first compared with returns; rolling signals then drive independent £100 accounts under single and split thresholds; a shared-capital comparison (Appendix E) tests reallocation and resizing; and top-N portfolios concentrate capital on the strongest recent signals.

1.5 Thesis Structure

Chapter 2 reviews the literature and identifies the research gap. Chapter 3 defines the data, sentiment measure, timing, portfolio rules and evaluation. Chapter 4 reports results from association to trading, Chapter 5 interprets them and Chapter 6 concludes. The appendices contain a text example, performance definitions, follow-up tests, supplementary examples and a capital-allocation comparison.

Chapter 2

Literature Review

The review considers why news might precede price changes, how financial tone is measured, and how textual measurements translate into decisions after costs and risk evaluation.

2.1 Why financial news may contain useful information

The efficient-market framework is the natural starting point. If public information is incorporated rapidly into prices, reading an already distributed headline should not systematically produce an excess return after trading costs (Fama, 1970). Investors may nonetheless differ in attention, interpretation or processing speed, so the empirical question is whether any predictable adjustment remains after publication and whether it is large enough to trade.

Tetlock (2007) finds that pessimism in a Wall Street Journal column predicts downward market pressure followed by reversal, and that unusually high or low pessimism accompanies higher trading volume. The reversal matters here: negative sentiment need not imply a permanently lower valuation, and a rule that exits on adverse news may miss the recovery if it stays out too long.

Heston and Sinha (2016), analysing more than 900,000 firm-level stories, find that daily news predicts returns for one to two days while weekly news contains information over longer intervals, and that negative information is incorporated more slowly. This motivates testing several lookbacks and separate buy and sell boundaries, although it cannot determine which window suits UK large-cap headlines scored by a different classifier.

News volume is a separate mechanism. More reports may draw attention to a development, but they can also be a consequence of a large price move, and syndicated repetition creates many observations without much independent information. Count is therefore analysed separately from mean tone.

Baker and Wurgler (2006) show that aggregate investor sentiment matters most for

stocks that are hard to value or arbitrage, which suggests that large, liquid FTSE constituents may respond less than the speculative stocks central to that mechanism.

The common implication is that measurement, timing and economic context should be specified together. A favourable article about an earnings surprise, an automated account of yesterday's gain and a broadly optimistic market comment can all receive positive labels while having different predictive content. A single cross-sectional coefficient averages these mechanisms. The present analysis therefore distinguishes contemporaneous association, subsequent returns and trading results, and uses industry composition and recurring selected names to interpret the differences.

2.2 Measuring financial language

Generic dictionaries misread financial language: a fall in costs is favourable and a fall in profits adverse. Loughran and McDonald (2011) show that many words classified as negative by a general dictionary are not negative in financial filings, motivating domain-specific measurement, although their evidence concerns filings rather than news.

VADER combines a lexicon with rules for emphasis and negation and was developed for social-media text (Hutto and Gilbert, 2014). It is a cheap reference, but handling negation is different from interpreting a financial comparison against expectations. BERT learns contextual representations from both sides of a token and can be fine-tuned for classification (Devlin et al., 2019), so the meaning of 'declined' in 'debt declined' depends on its context. Inputs have a finite token limit, and high classification probability does not guarantee calibrated certainty on a new data distribution. Araci (2019) adapts BERT to finance and evaluates it on Financial PhraseBank, whose labels express annotators' view of the positive, neutral or negative implication of a sentence for the company concerned (Malo et al., 2014). The labels represent human interpretation, not subsequent returns.

FinBERT's original notebook reports 85% accuracy and macro F1 of 0.83 on 969 held-out sentences (ProsusAI, 2019), while Araci reports approximately 86% accuracy in cross-validation. Neither measures trading profitability, and the two evaluation designs should not be conflated.

FinBERT's suitability here rests on domain fit, consistency and practicality: it returns the same three probabilities for every input, needs no parsing of generative answers, runs locally at archive scale and gives a directly inspectable positive-minus-negative score. The choice does not depend on claiming superiority over newer generative models.

2.3 From textual sentiment to returns

Text-based forecasting approaches combine language representations with price information (e.g. Halder, 2022; Zeng and Jiang, 2025), whereas Tetlock (2007) and Heston and

Sinha (2016) examine return relationships directly. Better classification does not establish a profitable trade, and a statistically significant association may be too small to survive costs, so a sentiment strategy must face price-only alternatives on common dates and execution conventions.

Simple threshold and ranking rules make the mapping from score to exposure explicit before introducing a flexible predictor (see also Yu et al., 2016). Separate entry and exit boundaries show how persistence affects turnover, and mean aggregation gives the signal a common range across companies. These choices sacrifice flexibility but ease economic interpretation.

A fixed rolling window weights eligible headlines equally and drops them abruptly at expiry. An exponentially decaying mean would smooth this boundary but add a parameter requiring independent selection, so window length and decision time are tested first. Headlines are concise and event-centred, whereas articles add explanation and background that can either correct or dilute a headline; with a finite model input, neither representation is ground truth (Appendix A).

Taken together, Araci (2019) supports finance-specific language measurement, while Tetlock (2007) and Heston and Sinha (2016) motivate horizon-dependent price tests. Their combined implication is conditional: an interpretable score can be relevant to financial information without retaining an exploitable advantage after publication. This project tests that distinction through progressively more selective portfolio rules. Weak forward associations consequently constrain the proposed use of sentiment, even if some concentrated portfolios perform well.

2.4 Costs and the evaluation of risk control

Midpoint execution assumes a trade inside the quoted spread, whereas a marketable purchase crosses to the ask and a sale to the bid; repeated adjustment can turn small favourable moves into net losses. Novy-Marx and Velikov (2016) show that trading costs eliminate much of the apparent profitability of high-turnover return anomalies.

Sharpe, Sortino and Calmar ratios add information that endpoint return omits, but sampling variation and serial dependence complicate their interpretation (Lo, 2002). Over six months they describe the observed path rather than estimate long-run performance.

Risk control requires more than a negative optimal threshold, which may simply imitate buy-and-hold. A benefit would require lower drawdown or downside deviation without an unacceptable loss of upside, relative to cash dilution or a simple price-based exit at similar average exposure.

Bailey et al. (2017) formalise the danger of selecting the best backtest from many configurations: searching thresholds, lookbacks, clocks and portfolio sizes makes a high maximum likely even without a persistent advantage, and Harvey et al. (2016) argue

that the significance hurdle should rise with the number of specifications tried. This is especially relevant to top-one and top-two portfolios, whose outcomes can change sharply when a single daily rank changes.

2.5 Research Gap and Contribution

The research gap is that evidence of sentiment-return association and improved text classification does not determine how a common headline score should change a portfolio after quoted trading costs. In particular, a signal may assist adverse-exposure control without ranking all future winners, while capital redistribution can erase a small apparent advantage. This study connects whole-period assessment, forward relationships, threshold decisions and relative sentiment selection within one fixed universe. It evaluates which conclusions survive bid/ask pricing and distinguishes market-index performance from beating each stock's own passive return.

Chapter 3

Data and Methodology

3.1 FTSE 100 Universe and Market Benchmark

The FTSE 100 comprises 100 large companies listed on the London Stock Exchange, screened for liquidity and investability by FTSE Russell and weighted by free-float market capitalisation; many constituents earn substantial revenue outside the UK (FTSE Russell, 2026).

Equation (3.1) gives the schematic weight, where P is price, Q shares outstanding, f the free-float factor and U_t index membership at time t; the benchmark is therefore not an equally weighted constituent portfolio.

$$w^I_{j,t} = \frac{P_{j,t}Q_{j,t}f_{j,t}}{\sum_{k\in U_t} P_{k,t}Q_{k,t}f_{k,t}} \tag{3.1}$$

The empirical universe contains the 100 names returned by the project's dated Refinitiv constituent-chain extraction. The file is therefore treated as a fixed research universe, not a fully reconstructed point-in-time membership history. The observed FTSE 100 index, whose membership and weights changed through the period, remains the market benchmark.

Broad ICB-style industry labels provide descriptive context. Table 3.1 reports each group's constituent count, mean whole-period price return and mean stock-linked headline count.

Energy has the highest mean return (16.70%) and coverage (9,217 headlines per constituent), Health Care the lowest return (−3.89%), and Financials are the largest group (27 constituents, 11.27%). Group sizes are small and headlines are not independent, so these differences are descriptive.

The London continuous trading session runs from 08:00 to 16:30 local time. Lookbacks measure elapsed hours, including eligible overnight and weekend news.

Figure 3.1 shows that the index rose 5.70% from the 2 January open to the 30 June

Broad ICB industry	Constituents	Mean return	Mean headlines
Financials	27	11.27%	5,543
Consumer Discretionary	16	−1.15%	3,566
Industrials	15	2.75%	4,812
Consumer Staples	10	0.39%	7,193
Basic Materials	8	7.31%	4,430
Health Care	5	−3.89%	6,126
Real Estate	5	7.12%	2,649
Utilities	5	7.51%	4,576
Energy	3	16.70%	9,217
Technology	3	2.35%	2,380
Telecommunications	3	−1.23%	5,505

Table 3.1: Fixed research universe by broad ICB-style industry, with mean constituent return and mean stock-linked headline count.

close, peaking in late February, falling sharply in March amid a Middle East and energy-price shock, and recovering in April as oil-price and ceasefire expectations improved (AJ Bell, 2026; Alliance News, 2026; Reuters, 2026). A long-only rule can appear protective in falling markets and lag during rapid rebounds.

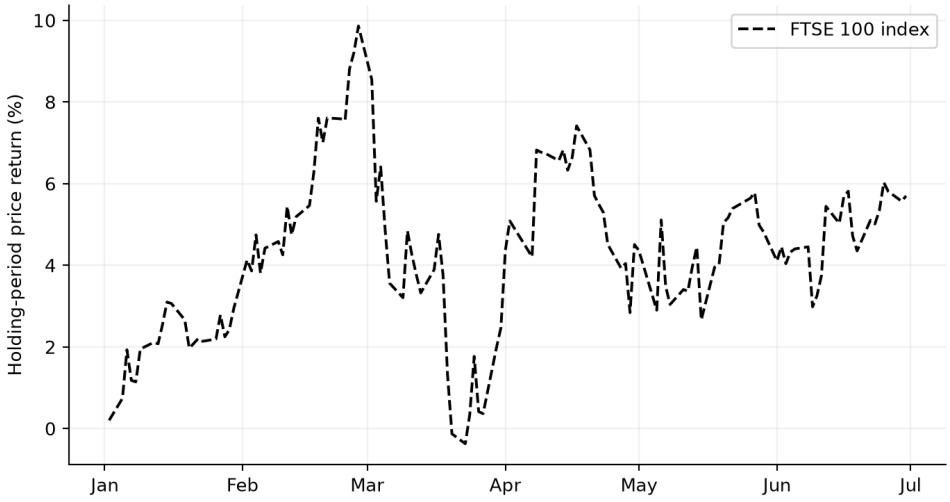


Figure 3.1: Observed FTSE 100 price-index return from the 2 January 2026 open to daily closes through 30 June. The final return is 5.70%; dividends are excluded. Source: Yahoo Finance (2026).

3.2 Model Selection and Measurement

The ProsusAI FinBERT checkpoint returns positive, neutral and negative probabilities summing to one. It receives cleaned text without a prompt or company parameter, so target ambiguity and domain transfer remain measurement limitations.

The published held-out FinBERT result is 85% accuracy, macro precision 0.81, macro recall 0.86, macro F1 0.83 and weighted F1 0.85 on 969 sentences (ProsusAI, 2019). Equa-

tion (3.2) defines class-specific precision and recall; Equation (3.3) defines F1. The exact published test-row assignment is unavailable, so alternative models are compared on 4,209 unique, unambiguous Financial PhraseBank sentences rather than claimed as like-for-like replications.

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c}, \qquad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c} \tag{3.2}$$

TP, FP and FN are true positives, false positives and false negatives; macro averages weight the classes equally and weighted F1 weights each class by its support. Opposite-direction errors (positive classified negative or vice versa) matter most for a threshold signal; this rate was not tabulated across models and is left for further work.

$$F1_c = \frac{2\,\text{Precision}_c\,\text{Recall}_c}{\text{Precision}_c + \text{Recall}_c}, \qquad F1_{\text{macro}} = \frac{1}{3}\sum_c F1_c \tag{3.3}$$

3.3 Data Compilation and Workflow

Refinitiv/LSEG supplies the English-language headline archive and prices: 504,320 stock-linked observations, 492,087 distinct stories and 2,077 source codes, the excess arising when one story is attached to several companies. Some items concern related but separately listed entities or several companies at once (Table 3.2; Appendix A), so part of a company signal may reflect news about other firms.

Daily OHLC data describe traded prices, and the intraday data summarises quoted bid and ask observations at five-minute intervals. Quotes are used as execution proxies, not evidence of completed trades.

Figure 3.2 shows how model selection, price data, strict-window sentiment and the trading simulations fit together.

The main strategy input excludes social-media source codes because their abbreviated, conversational form is less similar to Financial PhraseBank; the exclusion was also checked against strategy results on the same sample, so it is an additional in-sample researcher choice. This is a domain-consistency choice rather than a claim that social information is never valuable.

3.4 Headline Cleaning

Cleaning removes formatting and feed metadata (exchange and ticker tags, URLs, social handles, update labels, wire dates, duplicated price fragments and repeated HTML entities) while preserving meaning-bearing content such as gains, analyst actions, contract values and earnings comparisons.

Table 3.2 gives representative examples.

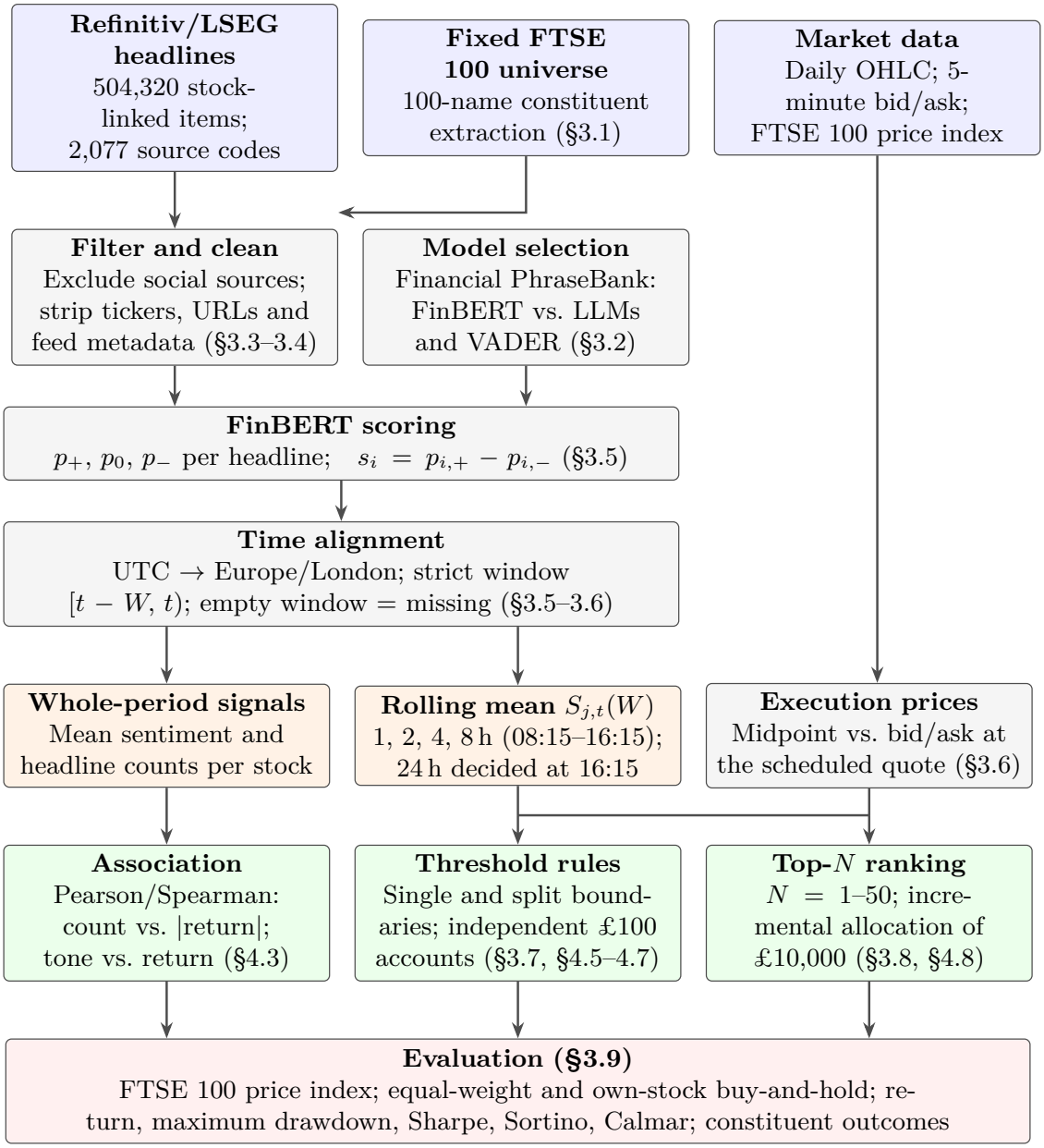


Figure 3.2: Research workflow, from data sources through text measurement, signal construction and the three analyses to economic evaluation. Section numbers show where each step is defined or reported.

Raw example	Cleaned input	Rule category
RECKITT <RKT.L>: BERNSTEIN RAISES TARGET PRICE TO 7500P FROM 6500P	RECKITT: BERNSTEIN RAISES TARGET PRICE TO 7500P FROM 6500P	Exchange ticker tag removed; financial figures retained
@YahooFinanceUK: Trending tickers: Micron, Sandisk Corporation, Ventyx Biosciences, BP and Reckitt [`https://t.co/example`]	Trending tickers: Micron, Sandisk Corporation, Ventyx Biosciences, BP and Reckitt	Social handle and URL removed
Reckitt Benckiser Group PLC (OTCMKTS:RBGLY) Given Consensus Rating of Buy	Reckitt Benckiser Group PLC Given Consensus Rating of Buy	Market identifier removed
Wednesday's Preview: Can Reckitt Benckiser (Bangladesh) (DSE:RECKITTBEN) continue its recent rise?	Wednesday's Preview: Can Reckitt Benckiser (Bangladesh) continue its recent rise?	Exchange identifier removed; country retained
Kingfisher hits year-high 15th time in three months; +0.3 GBX	Kingfisher hits year-high 15th time in three months	Redundant price suffix removed
UNITED KINGDOM LEISURE GOODS SECTOR BOTTOM 2 DAILY: TOP MOMENTUM DOWN IN THE PAST WEEK End-of-day: Monday January 19, 2026 16:00 (EST: 11:00)	UNITED KINGDOM LEISURE GOODS SECTOR BOTTOM 2 DAILY: TOP MOMENTUM DOWN IN THE PAST WEEK	Feed time metadata removed; direction retained
Weekly Summary: ICICI Prudential Life Insurance in 4th consecutive rise; +Rs24.35	ICICI Prudential Life Insurance in 4th consecutive rise	Summary prefix and currency suffix
Standard Chartered PLC - Transaction in own shares(with URL)	Standard Chartered PLC - Transaction in own shares	Link placeholder
Smith & Nephew wins contract	Smith & Nephew wins contract	Repeated HTML entity decoding

Table 3.2: Headline-cleaning rule categories with input and output examples. Financial comparisons and meaningful company qualifiers are retained.

Cleaning makes headlines more sentence-like without rewriting their economic direction; words are not stemmed and stop words are kept because they carry negation and comparison. Automated rankings and repeated price reports nonetheless remain different from hand-selected labelled sentences.

3.5 Sentiment Aggregation and Association

For cleaned headline i, Equation (3.4) converts class logits $z_{i,c}$ into softmax probabilities $p_{i,c}$ and defines the continuous net sentiment score s_i as positive probability minus negative probability. Neutral probability remains part of the normalising denominator, and the three class probabilities sum to one.

$$p_{i,c} = \frac{e^{z_{i,c}}}{\sum_{k\in\{+,0,-\}} e^{z_{i,k}}}, \qquad \sum_{c} p_{i,c} = 1, \qquad s_i = p_{i,+} - p_{i,-} \tag{3.4}$$

For company j, decision time t and lookback W, Equation (3.5) defines the eligible information set $\mathcal{I}_{j,t}(W)$ using an inclusive left boundary and exclusive cutoff. Equation (3.6) then defines the mean sentiment $S_{j,t}(W)$ over $N_{j,t}$ eligible headlines. If $N_{j,t}$ equals zero, the signal is missing rather than set to neutral.

$$\mathcal{I}_{j,t}(W) = \{i : j_i = j,\ t - W \le \tau_i < t\}, \qquad N_{j,t} = |\mathcal{I}_{j,t}(W)| \tag{3.5}$$

$$S_{j,t}(W) = \frac{1}{N_{j,t}} \sum_{i\in\mathcal{I}_{j,t}(W)} s_i, \qquad -1 \le S_{j,t} \le 1 \tag{3.6}$$

Every eligible headline enters the mean, including those whose highest probability is neutral: net scores of 0.80, -0.40 and 0.02 average 0.14. The mean keeps a common -1 to $+1$ scale across companies with different coverage, whereas a sum would confound tone with volume.

An empty window is missing information rather than neutral sentiment: a threshold account retains its position, and a top-N ranking excludes the company. A mean based on one headline can still move much more abruptly than one based on twenty.

The whole-period association uses the article-weighted mean over all eligible headlines for each company; because it uses information published after the start, it cannot select an initial investment prospectively.

Equation (3.7) defines Pearson correlation between variables X and Y; Spearman correlation applies the same calculation to ranks. Whole-period tests use constituent headline count or article-weighted mean sentiment as X and the stock's open-to-final-close return as Y. Prospective tests instead pair information fixed at the cutoff with later returns.

$$\rho_{X,Y} = \frac{\sum_j (X_j - \bar{X})(Y_j - \bar{Y})}{\sqrt{\sum_j (X_j - \bar{X})^2 \sum_j (Y_j - \bar{Y})^2}} \tag{3.7}$$

3.6 Decision Times and Execution Prices

The scheduled strategies make decisions every one, two, four or eight hours between 08:15 and 16:15 London time, with matching elapsed-hour lookbacks. The daily strategy uses a strict preceding 24-hour window and, for the principal results, decides at 16:15. Figure 3.3 introduces the local-time headline distribution used to choose that clock. 26.87% of linked observations arrive from 12:00 to 15:59 and another 4.34% from 16:00 to 16:59; 16:15 therefore captures the bulk of the afternoon news flow while leaving a 15-minute interval before the 16:30 continuous-session close. The 16:15 clock was motivated by this distribution; Table B.1 (Appendix B) shows that concentrated results at other clocks differ substantially.

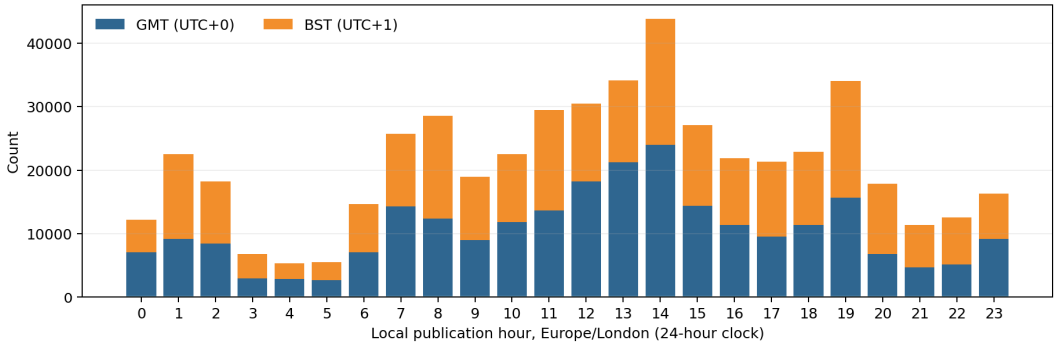


Figure 3.3: Distribution of stock-linked headline observations by hour.

Signals are calculated strictly before the scheduled London decision time. Only clean, non-crossed bid/ask observations with the exact scheduled five-minute label are eligible for execution. A missing price prevents a threshold trade. Ranking candidates require both a defined mean and an eligible price; a held stock without an execution quote remains held and receives its available end-of-day mark. News after the decision cutoff cannot enter the decision.

The intraday observation is a five-minute summary of quoted bid and ask prices, not a completed trade or a full limit-order-book reconstruction. Equation (3.8) defines the midpoint $m_{j,t}$ as the average of bid and ask. A 16:15 decision uses the quote interval labelled 16:15; final valuation uses the last clean quote at or before 16:25. This design prevents post-cutoff headlines entering the signal, but does not reproduce inference latency, queue position or market impact. Because the 16:15 quote summarises a five-minute interval, a headline arriving shortly before the cutoff could in principle be traded at a quote partly formed before it.

$$m_{j,t} = \frac{a_{j,t} + b_{j,t}}{2}, \qquad \text{spread}_{j,t} = a_{j,t} - b_{j,t} \geq 0 \tag{3.8}$$

An interval-summary quote is a historical approximation, and a delay can move the price either way. Midpoint execution is the spread-free reference; buying at ask and selling or valuing at bid incorporates the observed spread. The index uses official closes whereas portfolios use final continuous-session quotes, a small convention difference that is reported rather than interpreted.

3.7 Single and Split Threshold Rules

The single-threshold rule uses one boundary θ for both entry and exit. Equation (3.9) sets the intended holding state $h_{j,t}$ to one when an observed mean sentiment is at least θ and zero when it is below θ; if information is missing, the previous state is retained.

$$h_{j,t} = \begin{cases} \mathbf{1}(S_{j,t} \geq \theta), & N_{j,t} > 0 \\ h_{j,t-1}, & N_{j,t} = 0 \end{cases} \tag{3.9}$$

The grid spans -1 to $+1$. A boundary of -1 admits every score and so reproduces buy-and-hold; a negative selected threshold tolerates mildly adverse news and exits only under stronger negativity.

The split-threshold rule introduces buy boundary θ_B and sell boundary θ_S, constrained by θ_S no greater than θ_B. Equation (3.10) specifies the resulting hysteresis: a flat account enters at or above θ_B, a held account exits at or below θ_S, and intermediate values retain the prior state.

$$h_{j,t} = \begin{cases} 1, & h_{j,t-1} = 0,\ S_{j,t} \geq \theta_B \\ 0, & h_{j,t-1} = 1,\ S_{j,t} \leq \theta_S \\ h_{j,t-1}, & \text{otherwise} \end{cases}, \qquad \theta_S \leq \theta_B \tag{3.10}$$

Both rules are long-only. Each constituent initially has £100 in cash. A buy invests its available account balance, a sell converts the entire holding to cash, and later purchases use that balance. A sale at £120 therefore funds a £120 repurchase; a sale at £80 funds only £80. Gains and losses compound within each account, with no transfer between companies. The sum of 100 such accounts begins at £10,000 but is economically different from a freely reallocated portfolio.

C is cash, q share quantity and V marked account value; t^- denotes the state before trading, and midpoint experiments replace a and b by m.

$$q_{j,t} = \frac{C_{j,t^-}}{a_{j,t}} \text{ on entry}, \qquad C_{j,t} = q_{j,t^-}\, b_{j,t} \text{ on exit}, \qquad V_{j,t} = C_{j,t} + q_{j,t}\, b_{j,t} \tag{3.11}$$

3.8 Capital Allocation and Ranking

The ranking portfolios start with a shared £10,000. Incremental allocation keeps the quantity of every continuing holding unchanged, sells departing stocks and divides the proceeds equally among new entrants; with sales but no entrants, cash is retained. Weights therefore drift and the portfolio is not kept equally weighted, trading less resizing for different concentration. Appendix D gives a worked example.

$$L_t = H_{t-1} \setminus H_t, \qquad E_t = H_t \setminus H_{t-1}, \qquad A_t = \sum_{j \in L_t} q_{j,t^-}\, b_{j,t} \tag{3.12}$$

$$q_{j,t} = q_{j,t^-} \quad (j \in H_t \cap H_{t-1}), \qquad q_{j,t} = \frac{A_t}{|E_t|\, a_{j,t}} \quad (j \in E_t) \tag{3.13}$$

The top-N rule ranks the 24-hour mean sentiment and selects $N = 1, 2, 3, 4, 5, 10, 20$ or 50. Selection is relative: the best-ranked company can have a negative score if the rest of the eligible universe is more negative. Unless an additional absolute boundary is specified, a rank strategy should not be interpreted as a rule that invests only on optimistic days. The tested ranking portfolios use incremental allocation. Equal-weight redistribution provides a separate matched allocation comparison (Appendix E).

$$H_t^{(N)} = \text{Top}\, N_{j:\, N_{j,t}>0,\ \text{quote valid}}\ S_{j,t}(24\text{h}) \tag{3.14}$$

3.9 Benchmarks and Performance Measures

The common market benchmark is the observed FTSE 100 price index. It begins at the 2 January 2026 opening level of 9,931.30 and ends at the 30 June closing level of 10,497.10, a 5.70% price return. Daily closes trace the intervening path. It is not reconstructed by averaging constituent returns. The benchmark is constant across thresholds, portfolio sizes and decision clocks. An index-tracking investment would additionally involve product costs and tracking differences.

$$R_t^{\text{FTSE}} = 100 \left(\frac{I_t^{\text{close}}}{I_0^{\text{open}}} - 1 \right) \tag{3.15}$$

The FTSE series is a price-return index: dividends are excluded, consistent with the strategy price-return convention. This convention is not neutral for timing rules. The sample spans the main UK ex-dividend season, and a price-return rule that is out of a

stock over its ex-date records the mechanical ex-dividend fall as a loss avoided, although a holder would receive the dividend. Loss-avoidance and own-stock comparisons may therefore be flattened. A constituent-level buy-and-hold account remains necessary to answer whether timing a particular stock improves on owning that same stock. For this diagnostic, midpoint execution uses midpoint entry and final valuation; bid/ask execution buys at the first available clean ask and marks at the final clean bid. This passive entry need not coincide with a strategy's first decision time, so part of the measured difference reflects entry timing rather than the signal. These own-stock references are distinct from the aggregate FTSE 100 benchmark and are never relabelled as the index.

$$R_j^{BH} = 100\left(\frac{b_{j,T}}{a_{j,0}} - 1\right), \qquad R_T^{\text{strategy}} = 100\left(\frac{V_T}{V_0} - 1\right) \tag{3.16}$$

Daily returns use successive end-of-day values. Sharpe uses a zero risk-free rate and 252-day annualisation, Sortino a zero target, and Calmar annualised growth over absolute maximum drawdown (Appendix B); annualisation is a ratio convention, not a forecast.

$$r_d = \frac{V_d}{V_{d-1}} - 1, \qquad \text{Sharpe} = \sqrt{252}\,\frac{\bar{r}}{s_r} \tag{3.17}$$

$$\text{Sortino} = \sqrt{252}\,\frac{\bar{r}}{\sqrt{D^{-1}\sum_d \min(r_d, 0)^2}} \tag{3.18}$$

$$\text{MDD} = \min_d\left(\frac{V_d}{\max_{u\le d} V_u} - 1\right), \qquad \text{Calmar} = \frac{(V_D/V_0)^{252/D} - 1}{|\text{MDD}|} \tag{3.19}$$

For each independent account, profitability means a positive return and beating own-stock buy-and-hold means exceeding that stock's cost-matched passive return; Equation (3.20) cross-classifies the 100 accounts on both criteria. Shared portfolios are compared with the FTSE 100 at portfolio level.

$$n_{a,b} = \sum_{j=1}^{100} \mathbf{1}\left\{\mathbf{1}(R_j > 0) = a,\ \mathbf{1}(R_j > R_j^{BH}) = b\right\}, \qquad \sum_{a,b} n_{a,b} = 100 \tag{3.20}$$

No short selling, leverage, borrowing or interest on idle cash is assumed, and fractional units are permitted. The bid/ask experiment includes the observed quoted spread but excludes commissions, stamp duty, foreign-exchange conversion, dividends and market impact. Stamp duty (SDRT) of 0.5% on purchases of most UK-incorporated shares is typically several times the quoted spread of a FTSE 100 stock, so its omission matters most for high-turnover rules. As an illustration, the Top 1 portfolio makes approximately 111 full-capital purchases; at 0.5% each, terminal value would be multiplied by about $0.995^{111} \approx 0.57$, turning the reported 39.84% into a loss of roughly 20%. The correspond-

ing estimate for Top 2 is a loss of roughly 30%. These approximations ignore exempt non-UK-incorporated constituents, but they indicate that the concentrated results would not survive realistic UK costs at their current turnover.

3.10 Reliability and Researcher Choices

The analysis is an exploratory historical simulation. The researcher chooses the universe, source restriction, classifier, clock, parameter grid and evaluation measures; these choices influence the observed maximum. The search covers several thousand configurations (201 single thresholds across five schedules and two price modes, the split-threshold grid, and eight portfolio sizes at six decision clocks), and no deflated Sharpe ratio or probability-of-backtest-overfitting correction (Bailey and de Prado, 2014; Bailey et al., 2017) has been applied. Reliability is supported by a fixed checkpoint, cached inputs, explicit boundary rules and repeated calculations on the same observations. Construct validity remains limited because a sentence label measures perceived financial meaning rather than realised return. External validity requires later dates and other markets; it cannot be inferred from the number of correlated headlines alone.

Chapter 4

Results

4.1 Market Context

The FTSE 100 price index returns 5.70% over the study interval (Figure 3.1). This common reference differs from the equal-initial-capital collection of 100 stock accounts, whose equally weighted mean return is 4.97% (Table 3.1), so a strategy can beat many constituents' passive returns yet trail the capitalisation-weighted index. The market events described in Section 3.1 provide context but not causal attribution, which would require a separate event study with controls.

4.2 The input and the measurement choice

Across the full archive, headline classifications are 48.1% neutral, 32.7% positive and 19.2% negative. Neutral headlines are retained in the mean, which avoids amplifying the signal by averaging only directional items.

Coverage ranges from hundreds to tens of thousands of observations per constituent, so a high rank is not an equally precise estimate for every firm: one very positive headline can outrank many moderately positive ones.

Model	Test cases	Accuracy %	Macro P %	Macro R %	Macro F1 %	Weighted F1 %
FinBERT published test	969	85.00	81.00	86.00	83.00	85.00
Qwen 2.5	4,209	84.08	85.29	81.30	83.07	84.07
Gemma 2	4,209	80.21	78.96	86.88	81.13	80.44
Mistral	4,209	81.27	79.79	84.49	81.68	81.52
Llama 3.2	4,209	61.25	66.50	76.30	63.50	59.71
VADER	4,209	55.71	52.30	51.66	49.58	56.38

Table 4.1: Financial PhraseBank classification metrics. FinBERT uses the published 969-sentence held-out test; alternatives use available predictions on 4,209 unique matched sentences.

Table 4.1 does not rank FinBERT above the strongest generative alternative: Qwen 2.5 achieves a similar macro F1 (83.07% against 83.00%) on a different sample, and the

two evaluation designs cannot be ranked. The choice of FinBERT therefore rests on its stable probabilities, local inference at archive scale and reproducibility rather than superior accuracy. Excluding invalid generative outputs may flatten those models, and because FinBERT was fine-tuned on Financial PhraseBank, the benchmark cannot test transfer to the Refinitiv feed.

4.3 News quantity and average sentiment

Headline volume varies substantially by company. To prevent naturally high-coverage constituents dominating the comparison, each stock-day count is divided by that constituent's mean daily count. Figure 4.1 plots this normalised quantity against absolute open-to-close movement across 12,400 stock-days and introduces the fitted linear relationship.

The normalised relationship remains weak: Pearson r is 0.123 and Spearman ρ is 0.047, with the linear fit explaining about 1.5% of variation. The top volume quintile nevertheless has a higher mean absolute return (1.40%) than lower quintiles (about 1.08%-1.10%). News intensity therefore accompanies larger moves at the extreme without providing a strong standalone directional signal.

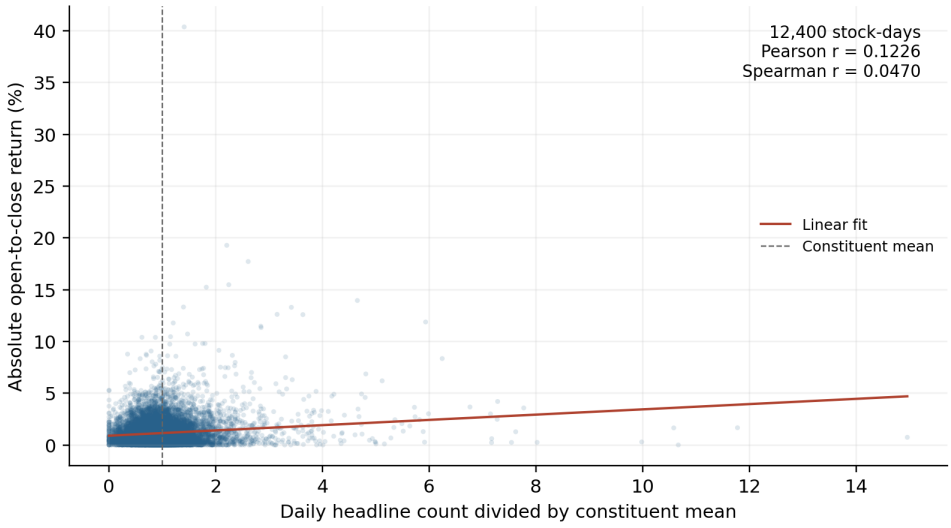


Figure 4.1: Constituent-normalised headline count against absolute daily open-to-close price movement over 12,400 stock-days. The red line is the linear fit; the dashed vertical line marks each constituent's own mean count.

The result is descriptive: it concerns the size of same-day movements, not their direction or a tradable return.

Figure 4.2 plots whole-period mean non-social sentiment against constituent return.

The whole-period relationship (Pearson $r = 0.436$ across 100 constituents; $t \approx 4.8$, although sector clustering and repeated coverage make the independence assumption behind that statistic optimistic) is stronger than the count association: companies with more favourable coverage tended to perform better over the same period. This supports

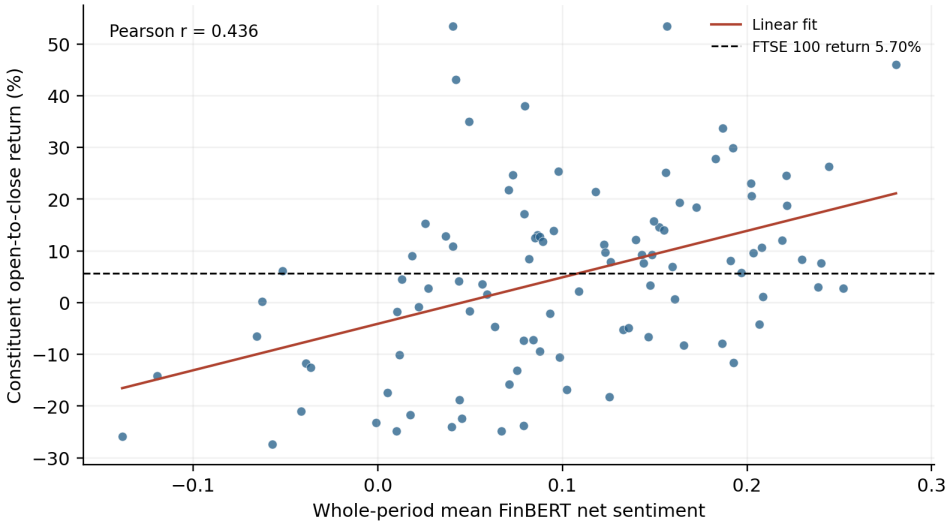


Figure 4.2: Whole-period mean non-social FinBERT sentiment against constituent open-to-close return. The red line is the linear least-squares fit; the dashed benchmark is the FTSE 100 price return of 5.70%.

average tone as an assessment variable but is equally consistent with successful firms attracting favourable coverage. Price-action headlines are retained (Section 3.3), so a rising stock accumulates favourable wording mechanically; re-estimating the correlation without automated price headlines, and within sectors, was not completed.

4.4 Predictive association across horizons

The short-horizon question is whether a signal ending at time t relates to the subsequent price change. This study does not report a formal forward-return test, such as a daily cross-sectional rank correlation between the 16:15 signal and the subsequent return, so conclusions about short-horizon prediction rest on the trading simulations in Sections 4.5 to 4.8. Same-period correlations are not reinterpreted as forward prediction.

A language model may summarise conditions accurately even when liquid-market prices already incorporate them, so a strong same-period association can coexist with weak forecasting.

A future test must account for overlapping windows, shared prices and common market shocks, using price-based controls and block-resampled uncertainty on a later period.

4.5 Single sentiment boundaries

Table 4.2 introduces the return-maximising single boundary for each schedule and execution convention. Midpoint rows are presented first, followed by bid/ask rows, so spread-free timing effects can be read before implementation costs.

The best 24-hour 16:15 boundary is -0.51 in both modes: midpoint returns 5.61% and

Window / clock	Execution	Threshold	Return	Profitable / 100	Beat own B&H / 100
1 hour	mid	−0.44	6.37%	66	53
2 hours	mid	−0.48	6.08%	62	53
4 hours	mid	−0.96	5.37%	65	42
8 hours	mid	−1.00	4.97%	64	22
24 hours, 16:15	mid	−0.51	5.61%	64	47
1 hour	bid/ask	−1.00	4.76%	62	29
2 hours	bid/ask	−1.00	4.76%	62	25
4 hours	bid/ask	−0.96	4.99%	64	39
8 hours	bid/ask	−1.00	4.87%	63	25
24 hours, 16:15	bid/ask	−0.51	5.31%	64	48

Table 4.2: Return-maximising single thresholds over -1 to +1 in 0.01 steps. Each constituent compounds an independent initial GBP100 account; the 24-hour strategy decides at 16:15. FTSE 100 price return: 5.70%.

bid/ask returns 5.31%, both slightly below the 5.70% index. They exceed the 4.97% mean return of the equally weighted constituents (Table 3.1), the more direct passive comparator for equal £100 accounts, although that figure is not cost matched. Several bid/ask optima lie at or near −1.00, a boundary that admits every score and therefore reproduces buy-and-hold: once the spread is paid, the best-performing rule at most schedules makes little or no use of sentiment. Figure 4.3 plots the full 24-hour response rather than only its maximum, showing that the selected negative boundary tolerates mildly adverse news and exits only under stronger negativity.

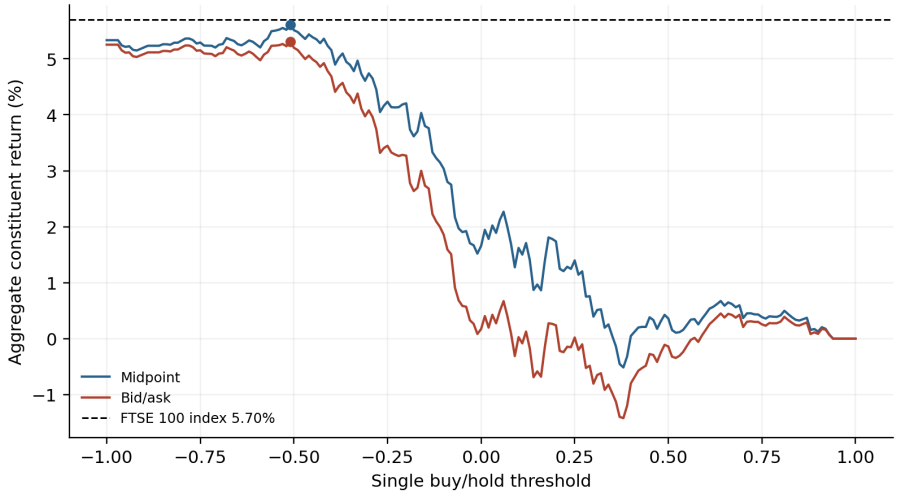


Figure 4.3: Single-threshold 24-hour strategy at 16:15 under midpoint and bid/ask execution. Points mark the in-sample return maxima; the dashed line is the FTSE 100 return.

The threshold curves are useful because the best point alone conceals whether neighbouring rules behave similarly. Broad plateaux indicate that a range of boundaries gives similar decisions, while narrow peaks suggest reliance on a small number of switches. The return function is not smooth in the mathematical sense: individual trades change when thresholds cross observed means. Lines joining grid points visualise the search and should not be interpreted as evidence of a differentiable response suitable for ordinary gradient

descent.

A selected negative threshold means that the rule tolerates neutral or mildly adverse coverage and exits only when sentiment is sufficiently unfavourable. This is compatible with an exposure-control interpretation. It is not proof of risk control, because the threshold was selected by return and can benefit from remaining invested during a rising period. The next comparisons therefore examine drawdown and which constituents benefit, instead of assuming that the sign of θ establishes the mechanism.

4.6 Separate entry and exit boundaries

Window / clock	Execution	Buy	Sell	Return	Beat own B&H / 100
1 hour	mid	0.00	−0.80	6.53%	54
2 hours	mid	0.00	0.00	5.91%	55
4 hours	mid	0.60	−0.90	5.45%	49
8 hours	mid	0.50	−1.00	5.18%	38
24 hours, 16:15	mid	−0.50	−0.50	5.51%	46
1 hour	bid/ask	−0.10	−1.00	4.91%	37
2 hours	bid/ask	0.10	−1.00	4.96%	40
4 hours	bid/ask	0.00	−1.00	5.06%	39
8 hours	bid/ask	0.50	−1.00	5.08%	40
24 hours, 16:15	bid/ask	−1.00	−1.00	5.25%	55

Table 4.3: Best split-threshold returns on the 0.1 grid, constrained by sell $\leq$ buy; midpoint rows first, bid/ask rows second. FTSE 100 price return: 5.70%.

For the 24-hour 16:15 rule, the best midpoint grid point is buy -0.50 and sell -0.50, returning 5.51%; the best bid/ask point is -1.00 for both boundaries, returning 5.25%. Figure 4.4 shows the full 16:15 return surfaces. At 24 hours the split-threshold maxima are slightly below the single-threshold maxima because the split grid is coarser (0.1 against 0.01), so the two searches are not nested. The bid/ask optimum of −1.00 for both boundaries is effectively buy-and-hold; that it nevertheless beats own-stock buy-and-hold for 55 constituents indicates that these counts partly reflect entry timing rather than the signal (Section 3.9).

Hysteresis lets a company remain held while its score moves between the boundaries, which can reduce spread crossings but also delays re-entry and may retain exposure through gradual deterioration.

4.7 Risk and constituent outcomes

This section compares return, drawdown and constituent outcomes. Table 4.4 first reports midpoint strategies, then bid/ask strategies, alongside the FTSE 100 benchmark; Figure 4.5 introduces their daily paths.

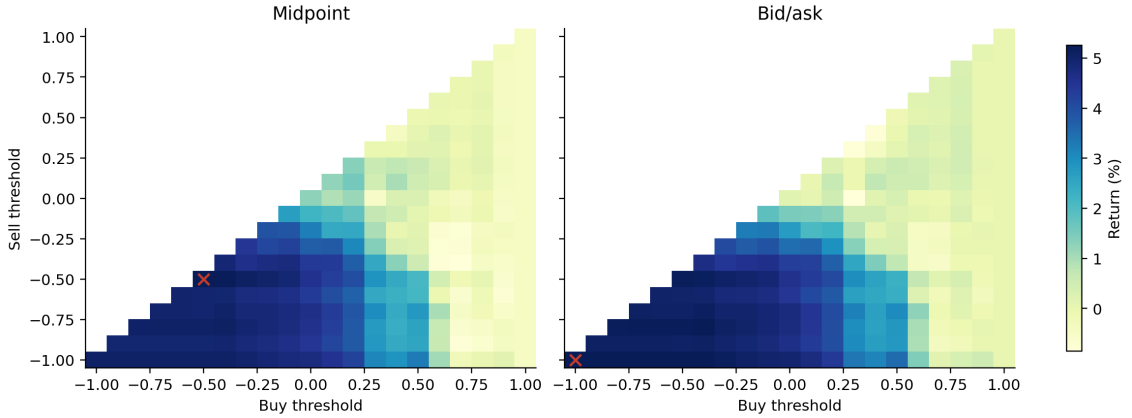


Figure 4.4: Split-threshold 24-hour return surfaces at 16:15. Darker cells indicate higher returns, red crosses mark maxima and blank cells violate sell threshold no greater than buy threshold.

Strategy	**Return**	**Maximum drawdown**	**Sharpe**	**Sortino**	**Calmar**
FTSE 100 index	5.70%	−9.32%	0.89	1.29	1.28
1 hour mid	6.37%	−8.83%	1.05	1.59	1.51
24 hours, 16:15 mid	5.61%	−10.65%	0.85	1.25	1.10
1 hour bid/ask	4.76%	−10.77%	0.72	1.06	0.92
24 hours, 16:15 bid/ask	5.31%	−10.70%	0.81	1.19	1.04
Top 1 at 16:15	39.84%	−17.90%	1.94	3.00	5.46
Top 2 at 16:15	19.88%	−7.69%	1.51	2.39	5.79
Top 3 at 16:15	3.43%	−13.56%	0.40	0.61	0.52
Top 4 at 16:15	8.52%	−13.93%	0.85	1.30	1.30
Top 5 at 16:15	5.42%	−11.32%	0.62	0.98	1.00
Top 10 at 16:15	0.66%	−12.36%	0.16	0.24	0.11
Top 20 at 16:15	−1.06%	−12.52%	−0.06	−0.08	−0.17
Top 50 at 16:15	1.59%	−12.53%	0.29	0.40	0.26

Table 4.4: Return and risk diagnostics. Ratios use daily portfolio values, a zero target and 252-day annualisation; the 24-hour threshold rows decide at 16:15.

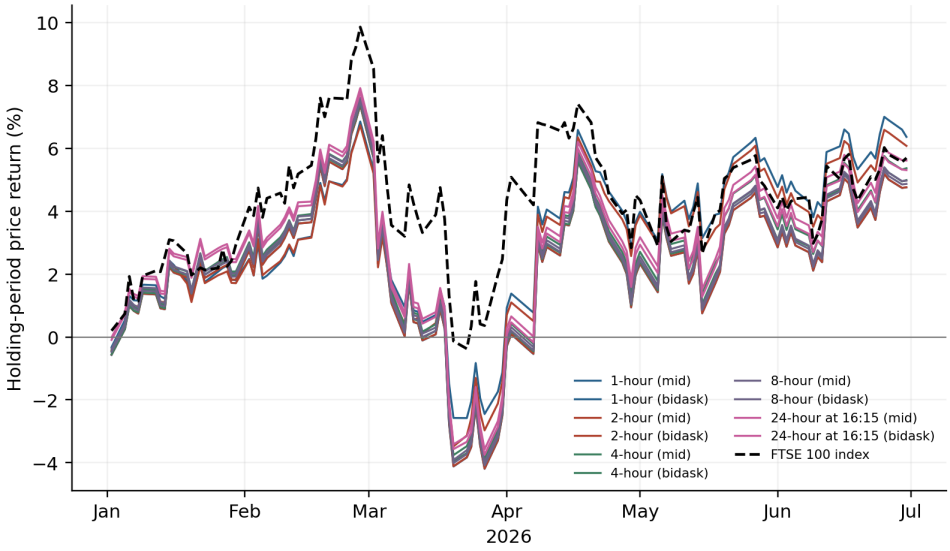


Figure 4.5: Daily paths of the best-return single-threshold configuration for each schedule. The 24-hour series decides at 16:15; the dashed line is the FTSE 100 price index.

The 24-hour 16:15 midpoint rule returns 5.61% with a 10.65% maximum drawdown, Sharpe 0.85, Sortino 1.25 and Calmar 1.10. Bid/ask execution reduces return to 5.31% with similar drawdown and ratios. Both drawdowns are deeper than the index drawdown of 9.32%. With roughly six months of daily data, the standard error of an annualised Sharpe ratio is approximately 1.5 for ratios near 0.9 and 2.4 for the Top 1 ratio of 1.94 (Lo, 2002), so none of the ratio differences in Table 4.4 is statistically distinguishable. At constituent level, the strategy most improves on passive ownership for Endeavour Mining (+19.74 percentage points), Antofagasta (+16.99) and Babcock (+10.67), while it lags Computacenter (-11.68), Howden Joinery (-10.53) and Fresnillo (-10.06). Figure 4.6 shows when these stocks were actually held. Three of these six extremes are metals miners, appearing on both sides of the comparison, which suggests that stock volatility rather than signal quality drives the largest deviations from buy-and-hold.

Execution	Profitable	Beat own B&H	Constituents
mid	Yes	Yes	35
mid	Yes	No	31
mid	No	Yes	18
mid	No	No	16
bid/ask	Yes	Yes	30
bid/ask	Yes	No	34
bid/ask	No	Yes	18
bid/ask	No	No	18

Table 4.5: Mutually exclusive constituent outcomes for the best-return hourly midpoint and 24-hour 16:15 bid/ask single-threshold strategies. Each mode totals 100 constituents; own-stock passive returns are cost matched.

Table 4.5 separates profitable strategies that beat or lag each stock's passive return

from loss-making strategies that nevertheless reduced or worsened the loss. The outcomes are nearly symmetric: under bid/ask execution, 18 loss-making accounts improve on passive ownership and 18 do worse. Without a placebo using randomly timed exits, this pattern cannot be distinguished from timing volatile stocks at random. Table 4.5 also compares two different rules, so its rows should not be read as a like-for-like execution comparison. Small sector groups and common market shocks prevent these comparisons being treated as independent sector effects.

The 24-hour bid/ask rule illustrates both sides. Babcock loses under the strategy but loses materially less than passive ownership, while Computacenter remains profitable but gives up part of a very strong passive gain. The evidence therefore supports sentiment as a candidate exposure overlay more clearly than as a universal return-maximising selector.

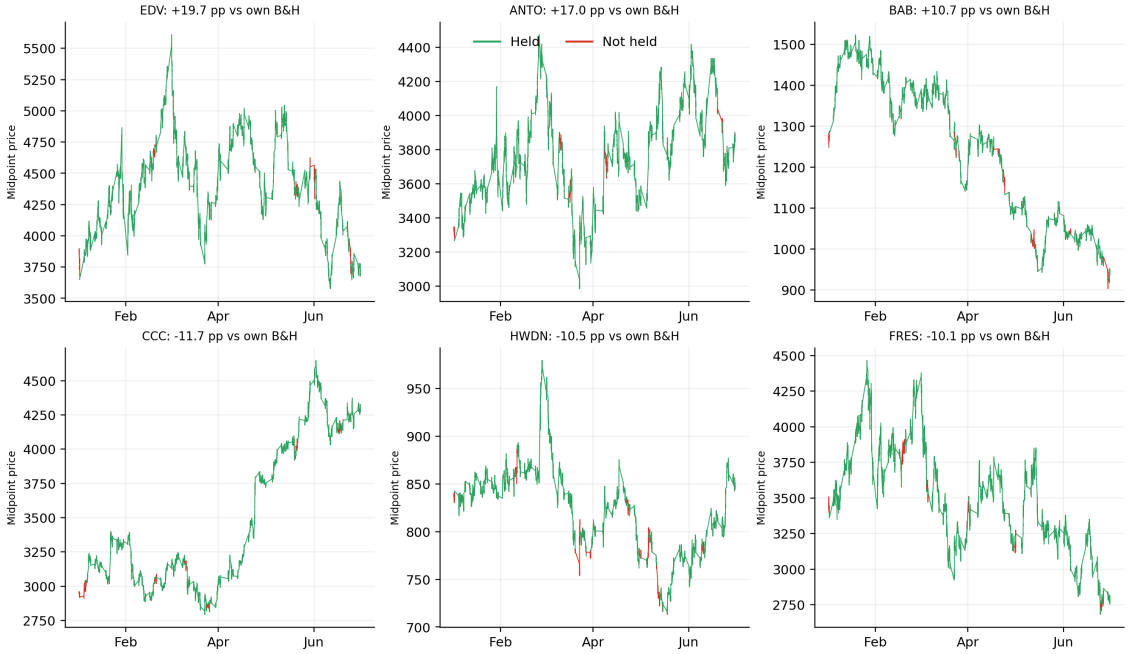


Figure 4.6: Midpoint price histories for the greatest overperformers and underperformers against own-stock buy-and-hold under the 24-hour 16:15 bid/ask rule. Green segments are held; red segments are not held.

4.8 Ranking stocks by recent sentiment

This section turns from thresholds to relative selection. Table 4.6 introduces daily 16:15 Top-N portfolios ranked by the preceding 24-hour mean sentiment under bid/ask execution and incremental funding.

At 16:15, Top 1 returns 39.84% and Top 2 19.88%, but performance is highly concentrated and timing sensitive. Figure 4.7 therefore shows only Top 1 through Top 5 against the FTSE 100, keeping the principal comparison legible; broader Top 10, 20 and 50 outcomes remain in Table 4.6. Returns are not monotonic in N, and the Top 10, 20 and 50

Top N	Return	Excess over index pp	Stocks used	Order events	Maximum drawdown
1	39.84%	+34.14	44	221	−17.90%
2	19.88%	+14.18	62	434	−7.69%
3	3.43%	−2.27	73	627	−13.56%
4	8.52%	+2.82	75	826	−13.93%
5	5.42%	−0.27	81	991	−11.32%
10	0.66%	−5.04	97	1822	−12.36%
20	−1.06%	−6.76	98	3034	−12.52%
50	1.59%	−4.11	100	4642	−12.53%

Table 4.6: Daily 16:15 top-N portfolios using preceding 24-hour mean sentiment, bid/ask execution and incremental funding. Continuing quantities remain unchanged. All returns are holding-period price returns; the market-index reference is 5.70%.

portfolios all fall below the 4.97% equal-weighted constituent mean, so sentiment ranking adds value only in the most concentrated portfolios. Table B.1 (Appendix B) shows that the concentrated result is also highly sensitive to the decision clock: Top 1 returns between −12.3% and −12.9% at 11:15 to 13:15, but 40.15% at 15:15 and 39.84% at 16:15. These results exclude stamp duty, which on the illustrative estimate in Section 3.9 would turn both Top 1 and Top 2 into losses.

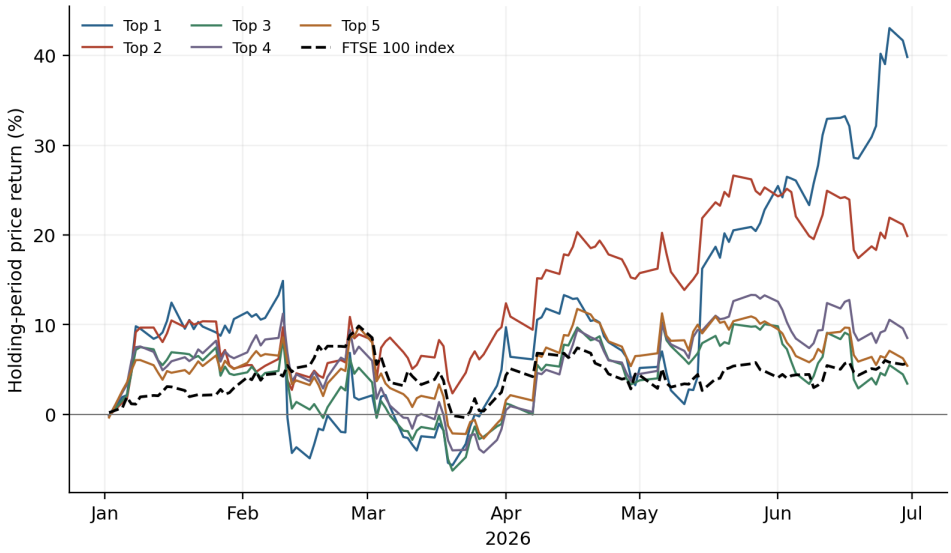


Figure 4.7: Daily combined returns for Top 1 to Top 5 sentiment portfolios at 16:15 using bid/ask execution and incremental funding. The dashed line is the FTSE 100 index.

Table 4.7 introduces the concentration behind the headline result. Computacenter is selected on 22 days and Tritax Big Box REIT on 16. The largest listed contributions come from Hiscox, Games Workshop and Tritax Big Box REIT, while ICG detracts. Frequent constituents span Industrials, Financials and Real Estate rather than one uniform sector, and several have strong passive returns; the strategy's success is therefore partly a concentrated stock-selection result rather than broad evidence that sentiment ranks the entire universe.

Whole-period mean sentiment in Table 4.7 is descriptive and is not used to select

Ticker	Company	Days	Mean sentiment	Passive return	P&L contribution
CCC	Computacenter PLC	22	0.281	46.07%	£915
BBOX	Tritax Big Box Reit PLC	16	0.144	7.67%	£1,013
GAW	Games Workshop Group PLC	7	0.095	13.90%	£1,062
SSE	SSE PLC	5	0.219	12.10%	£425
CCEP	Coca-Cola Europacific Partners PLC	5	0.123	9.71%	£135
HSX	Hiscox Ltd	4	0.192	29.89%	£1,290
HLMA	Halma PLC	4	0.208	10.63%	£166
ICG	ICG PLC	4	0.102	−16.88%	−£251
AAF	Airtel Africa PLC	4	0.187	−7.93%	£383
ADM	Admiral Group PLC	3	0.123	11.20%	£9

Table 4.7: Ten most frequently selected constituents in the 16:15 Top 1 portfolio, with whole-period mean sentiment, passive return and total marked-to-market P&L contribution.

holdings. P&L contribution includes realised sales plus the final bid-valued open position.

The principal portfolio comparator is the observed FTSE 100 price index. Each constituent's own passive return remains a separate timing diagnostic, and a price-return ranking with identical dates and execution rules is the appropriate additional control for momentum (Jegadeesh and Titman, 1993); this control and a randomised-signal placebo were not completed. A static full-period sentiment selection uses future information and cannot be interpreted as an investable initial allocation. Further comparisons should freeze all these definitions before evaluating later dates.

Chapter 5

Discussion

5.1 What it means for sentiment to be useful

The results distinguish assessment from prediction. Whole-period mean sentiment has a positive cross-sectional association with constituent return, while normalised news volume has only a weak relationship with absolute daily movement. At shorter horizons, a headline score is noisy, target ambiguity remains and the market may incorporate information before the next scheduled quote. Sentiment is therefore informative about company conditions without being established as a general next-period return predictor.

The association is still a useful contribution. It demonstrates that the language measurement is not unrelated noise when aggregated at company level, and that quantity alone does not provide the same explanatory pattern. In relation to Tetlock (2007) and Heston and Sinha (2016), the result adds evidence from a different market, period, news mixture and classifier. It should be described as an empirical extension or replication of a broad sentiment-return relationship.

The contrast across horizons also helps explain why trading results need not match a correlation coefficient. A threshold rule is nonlinear and state dependent: it responds only when a boundary changes the position. A rank rule compares companies and can invest even when all eligible scores are negative. Both can generate a return pattern different from a linear pooled relationship. Conversely, a profitable backtest does not prove that the underlying score has broad predictive power. It may capture a small number of favourable selections, a particular regime or an allocation effect.

5.2 Evidence for an exposure overlay

The risk statistics show why terminal return alone is insufficient. Only the hourly midpoint threshold configuration reduces drawdown relative to the index (-8.83% against -9.32%), and this advantage disappears under bid/ask execution, whereas Top 1 combines a much

larger gain with a drawdown near 18%. The bid/ask threshold maxima remain below the market index. This supports a narrower hypothesis: adverse sentiment might help control exposure when combined with another selection rule. Demonstrating that benefit requires comparison with passive or price-based controls at similar average exposure.

An exposure-control overlay is a rule that reduces or closes a position selected by another strategy when an adverse condition is met. For example, a value or momentum portfolio could retain its selection process but suspend a holding when mean news sentiment crosses a predeclared negative boundary. Re-entry and turnover limits must also be specified. Its usefulness would be a better downside outcome for the combined strategy, not merely a positive return from the sentiment rule in isolation.

A credible test would compare the base strategy, the same strategy with a sentiment exit, and equally simple price-based or volatility-based exits. Each must have matched capital, execution, dividend and tax treatment. A cash-weighted passive portfolio with similar average exposure would help determine whether smaller drawdown is merely the consequence of holding less stock. Incremental benefit would be supported if sentiment improves downside outcomes beyond those controls without an excessive loss of upside or disproportionate trading cost.

This proposed use fits the asymmetry observed in parts of the literature. Heston and Sinha's slower incorporation of adverse information suggests a reason to examine negative signals separately, while Tetlock's reversal evidence cautions against remaining out indefinitely after pessimistic coverage. Separate thresholds offer a simple way to formalise persistence and recovery, but their interpretation depends on measured outcomes. The project provides motivation for further testing, not a completed proof that sentiment should control risk in a live portfolio.

5.3 Concentration and decision time

The top-N results are sensitive to portfolio size and decision time. Smaller N gives one company and one holding interval greater influence over the whole portfolio. Heston and Sinha (2016) motivate testing information horizons, but do not establish that one particular clock should dominate. Bailey et al. (2017) explain why selecting the best time from many backtests can exaggerate the apparent advantage. The present concentration results are therefore hypotheses for independent testing rather than broad confirmation of sentiment ranking.

The daily window changes the information set as well as the execution price. A 24-hour window ending at 16:15 includes that afternoon's news and removes the preceding afternoon's older news. This can change the rank of a company even if its price hardly moves over the intervening hours. In a top-two portfolio, changing one name replaces a substantial portion of capital. The return difference between two windows therefore

cannot be interpreted as simply the profit made during the five hours separating them.

Selection frequency and industry exposure clarify the mechanism further. Concentrated strategies can repeatedly choose companies in a favourable sector and favour automated coverage with a consistent positive vocabulary. Neither interpretation automatically implies a language-model failure. It does mean that the strategy may be partly a sector or source exposure. A useful extension would compare within-industry ranks and measure whether sentiment still selects favourable returns after controlling for momentum, volatility, market sensitivity and coverage structure.

The top-N sensitivity is a reason to report the full surface of outcomes. A strong top-one result alongside weak top-five or top-ten results is evidence about concentration, not broad confirmation that sentiment ranks the entire cross-section correctly. If an effect is persistent, it should survive reasonable changes in decision time and portfolio size on later observations. The research identifies configurations worth testing, while preserving the distinction between selecting a promising hypothesis and establishing its out-of-sample performance.

5.4 Why execution changes the conclusion

Midpoint execution is a spread-free reference, while bid/ask execution pays the quoted spread. The grid shows that midpoint and bid/ask maxima can occur at different boundaries and schedules. Fama's (1970) efficiency argument concerns opportunities after information and implementation costs; the observed loss of advantage when crossing the spread is consistent with that distinction. The matched allocation experiment (Appendix E) additionally shows that lower turnover can help without guaranteeing index outperformance. No universal ranking of lookback length follows. At the 24-hour horizon the spread costs only about 0.3 percentage points and does not change the conclusion; the omitted stamp duty is the larger cost for UK equities (Section 3.9).

Retaining continuing holdings avoids trading on small target-weight changes but makes weights path dependent, can increase concentration and can leave cash idle; equal weighting diversifies more but requires more resizing.

Five-minute summary quotes omit depth and the exact price sequence, and fractional shares may not be available, so the bid/ask results are spread-aware historical simulations rather than evidence of achievable execution.

5.5 Interpretation and limits

A high FinBERT probability does not certify an economic forecast, and cleaning cannot resolve every ambiguous comparison or retrospective price report; Appendix A shows opposite headline and article classifications arising from different emphasis.

The six-month sample limits the range of market regimes, and sector dependence and repeated stories reduce the effective number of independent observations, so inference should preserve market-wide and temporal dependence.

Parameter selection is another limit. Thresholds, source filters, window lengths, decision times and stock counts were all examined with the same return history. This creates optimism in selected maxima even when each underlying calculation is correct. The study consequently reports them as descriptive sample outcomes. No claim of reliable prediction rests on the largest number alone. The next stage should predeclare a small set of rules and compare them on later dates without revisiting the selection criteria. Two further limitations apply to every trading result: returns exclude dividends during the main ex-dividend season, and stamp duty is not charged. Each is likely to flatten the timing and concentrated-selection results, and correcting them is a precondition for any practical claim.

5.6 Further investigation

Further research should resolve specific limits. An untouched later period would test whether the threshold and top-N findings persist after parameter selection; matched price-only and cash-exposure controls would test whether loss avoidance is attributable to sentiment. Re-estimating all results with dividend-adjusted prices and stamp duty, and comparing them with a randomised-signal placebo, is the most immediate step. A new target-feed annotation set would test model transfer independently of PhraseBank. These steps matter because they distinguish a useful addition to an existing investment process from a pattern selected retrospectively from many possibilities.

An event-level analysis would help distinguish retrospective price reports from new fundamental information. Company announcements, analyst revisions and contract news could be compared with automated performance summaries, retaining the original wording and publication times. Deduplicating syndicated stories would test whether repeated coverage amplifies a narrative without adding independent information. These experiments would be more informative than repeatedly searching the same threshold grid without changing the evidence available.

Recency weighting is another justified extension. An exponentially weighted average would allow the contribution of an article to decline gradually rather than disappearing at a hard boundary. The half-life should be selected without reference to the final test returns, and an unweighted mean should remain the comparator. A minimum effective article count or uncertainty estimate could limit decisions dominated by one observation. These additions should be evaluated for incremental predictive and risk-control value, not adopted merely because they make the signal more complex.

Finally, paper trading would connect the historical experiment to observable execution.

It should record when a headline was received, when its score was available, the proposed order, the contemporaneous bid and ask and the simulated fill. Comparing those records with the interval-summary backtest would quantify timing and spread assumptions. This would allow the project to progress from demonstrating a plausible research use of sentiment toward assessing a controlled application alongside other investment rules.

Chapter 6

Conclusion

This dissertation asks whether financial-news sentiment is useful for assessing or predicting FTSE 100 constituent returns and for informing long-only investment decisions, using FinBERT as a measurement tool linked to economic evaluation through strict information windows, quoted execution prices and passive benchmarks.

Whole-period mean sentiment is positively associated with constituent return, whereas constituent-normalised headline volume explains little daily absolute movement. The best 24-hour 16:15 single-boundary rules return 5.61% at midpoint and 5.31% at bid/ask, below the FTSE 100's 5.70%. Both exceed the 4.97% equal-weighted constituent mean, the more direct passive comparator. Their constituent outcomes show loss avoidance for some weak stocks and sacrificed upside for some winners, in nearly equal measure. Concentrated Top 1 and Top 2 portfolios outperform strongly in sample, but are sensitive to timing, turnover and a small set of names; on an illustrative estimate, stamp duty would turn both into losses.

The thesis question therefore receives a qualified answer. Financial-news sentiment is useful as a descriptive measure and as an interpretable candidate for exposure control and concentrated selection. The evidence does not establish a general predictive edge across stocks or horizons. The fixed-universe membership mismatch, parameter search, interval-summary execution, omission of stamp duty and dividends, the absence of a formal forward-return test and the absence of an independent test period further constrain causal or deployable claims.

The most defensible practical direction is to test sentiment as a complementary signal, particularly for exposure control, with concentrated ranking treated as a separate selection hypothesis. The project provides evidence and specific mechanisms for that further work. Independent dates, matched price-based controls and recorded execution are required to establish whether the observed benefits persist beyond the sample and add value to an existing investment process.

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Appendix A

Headline and Scored Article Excerpt

This example is drawn from the broader company archive and illustrates input sensitivity; Nvidia is not a constituent of the FTSE sample. The text below is the excerpt actually supplied to the classifier. The original publisher's full article is not claimed to have been scored. The negative daily-price headline contrasts with positive expectations discussed later in the excerpt.

Headline

Nvidia (NVDA) Stock Dips While Market Gains: Key Facts

Cleaned headline supplied to FinBERT

Nvidia Stock Dips While Market Gains: Key Facts

Headline classification

negative (positive 0.022, neutral 0.118, negative 0.860)

Article excerpt supplied to FinBERT

For best results when printing this announcement, please click on link below: `https://newsfile.refinitiv.com/getnewsfile/v1/story?guid=urn:newsml:reuters.com:20260108:nNRAz2kvum&default-theme=true` Jan 08, 2026 Nvidia (NVDA) ended the recent trading session at $185.00, demonstrating a -2.17% change from the preceding day's closing price. The stock's performance was behind the S&P 500's daily gain of 0.01%. Shares of the maker of graphics chips for gaming and artificial intelligence have appreciated by 2.9% over the course of the past month, outperforming the Computer and Technology sector's loss of 0.69%, and the S&P 500's gain of 0.86%. The investment community will be closely monitoring the performance of Nvidia in its forthcoming earnings report. On that day, Nvidia is projected to report earnings of $1.52 per share, which would represent year-over-year growth of 70.79%. Alongside, our most recent consensus estimate is anticipating revenue of $65.49 billion, indicating a 66.5% upward movement from the same quarter last year. Looking at the full year, the Zacks Consensus Estimates suggest analysts are

expecting earnings of $4.66 per share and revenue of $211.91 billion. Investors should also note any recent changes to analyst estimates for Nvidia.

Article classification

positive (positive 0.920, neutral 0.018, negative 0.063)

Appendix B

Performance and Portfolio Definitions

Holding-period return is final portfolio value divided by initial capital minus one. It is not annualised and excludes dividends. The common market comparison is the observed FTSE 100 price index. Portfolio risk ratios use successive final continuous-session marks and initial cash before the first investment, retaining the first day's spread loss. Index risk ratios use daily index closes, so the closing convention is explicit.

Maximum drawdown is the minimum of the current value divided by the highest preceding value minus one. The daily return sample standard deviation uses one degree of freedom. The Sortino denominator includes a zero downside contribution for each non-negative day. A zero denominator is reported as undefined rather than replaced by an arbitrary large number. Calmar uses 252 divided by the number of daily return observations as the growth annualisation exponent.

For independent accounts, beating own-stock buy-and-hold means ending above that constituent's cost-matched £100 passive account. For shared capital, performance is compared with the FTSE 100 price index. Per-company raw profit in a reallocated portfolio depends on capital allocation and holding time, so it is not an estimate of the return from allocating £100 to that stock for the whole period.

Table B.1 introduces decision-time sensitivity for Top-N portfolios. The accompanying Figure B.1 then isolates the Top 1 and Top 2 results across clocks, showing that concentrated performance is strongest late in the session.

London time	Top 1	Top 2	Top 5	Top 10	Top 20	Top 50
11:15	−12.28%	−6.38%	−2.94%	−4.32%	−4.86%	2.10%
12:15	−12.87%	−7.87%	−2.05%	−2.32%	−4.40%	−0.35%
13:15	−12.81%	5.84%	5.10%	3.65%	1.19%	−1.12%
14:15	23.08%	11.04%	9.30%	−2.69%	0.44%	−1.41%
15:15	40.15%	21.82%	1.24%	−0.98%	−1.64%	0.33%
16:15	39.84%	19.88%	5.42%	0.66%	−1.06%	1.59%

Table B.1: Decision-time sensitivity of incremental bid/ask Top-N returns. Each row uses a strict preceding 24-hour window and a common end-of-day portfolio mark; the FTSE 100 return is 5.70% at every clock.

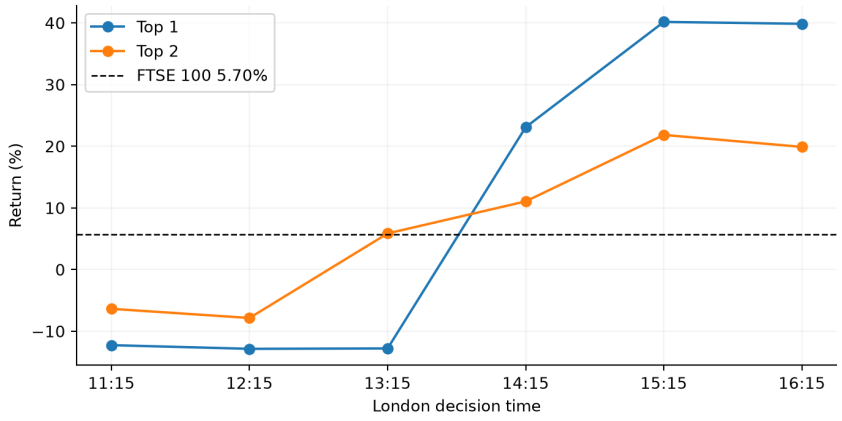


Figure B.1: Holding-period Top 1 and Top 2 returns by London decision time using a strict preceding 24-hour window. The dashed benchmark is the FTSE 100 return.

Appendix C

Research Interpretation and Follow Up Tests

The descriptive association, a forward-return test and the trading experiment answer different questions. The whole-period correlation supports association. A lagged correlation with future return supports predictive association under its timing definition. An investable excess return requires a decision rule and achievable prices after all relevant costs. Evidence at one level should not be substituted for evidence at another.

The proposed risk overlay should be evaluated against both the unmodified base strategy and controls with similar average market exposure. Relevant endpoints include maximum drawdown, worst daily loss, downside capture, expected shortfall, turnover and recovery time, as well as total return and the ratios reported here. The held-out period should include different market conditions, and inference should preserve common daily shocks across constituents.

The positive whole-period association is a contribution in the sense of a documented result for this market, corpus and period. Prior literature already establishes related sentiment-return relationships, so no claim of first discovery is made. The stronger contribution is connecting that finding to the conditions under which sentiment does or does not improve an explicit investment decision.

Appendix D

Supplementary Examples

Consider an illustrative headline reporting that a company gained over the preceding week. FinBERT may correctly label its wording positive, but the gain has already occurred. Its classification therefore supplies evidence about the reported event, not proof of a future increase. Conversely, sustained adverse operating news may justify investigating a loss-avoidance rule even when the next five-minute return remains difficult to forecast. The empirical design keeps these interpretations separate.

As an example of incremental allocation, suppose a portfolio holds A and B, and the new selected set is B and C. A is sold at its bid, B's quantity is unchanged, and all released capital purchases C at its ask. If the old set is A, B and C and the new set is B, D and E, A and C are sold before the available proceeds are divided equally between D and E. B is untouched. This rule implements the practical idea that sentiment changes the membership of the portfolio without repeatedly rebuilding holdings that remain eligible.

Appendix E

Capital-Allocation Comparison

This appendix reports the matched capital-allocation comparison, which tests whether sharing capital across constituents changes the threshold-rule results. The ranking portfolios in Section 4.8 use incremental allocation only (Section 3.8).

The shared-capital experiments start with £10,000. Full equal-weight allocation computes the desired portfolio after all signals have been assessed, then trades only the net differences needed to reach the target weights. Simultaneous exits and entries are considered together. Proceeds from exits can increase continuing positions, and introducing a new constituent can require partial sales elsewhere. The cost calculation accounts for buying at ask and selling at bid so that the target allocation does not imply borrowing money to pay the spread.

H_t is the target set, Δq is target minus existing quantity, and x_t is the common midpoint value per selected stock solved subject to available cash and the spread. All desired quantities are computed together before executing net orders.

$$q^*_{j,t} = \frac{x_t}{m_{j,t}} \mathbf{1}(j \in H_t), \qquad C_t = C_{t^-} - \sum_j \left[a_{j,t} (\Delta q_j)^+ - b_{j,t} (-\Delta q_j)^+ \right] \geq 0 \tag{E.1}$$

The matched allocation comparison fixes the signal (a strict 24-hour mean at 16:15, entry at or above -0.20 and exit at or below -0.90) and compares independent £100 accounts, a shared £10,000 portfolio resized to equal midpoint-valued holdings, and incremental funding. These boundaries are illustrative, not allocation-specific optima.

Table E.1 introduces the matched capital-allocation experiment. The same 24-hour 16:15 signal and -0.20/-0.90 boundaries are held fixed while accounting changes from independent GBP100 accounts to shared equal-resizing and incremental portfolios.

Under bid/ask execution, independent accounts return 4.90%, equal resizing returns 4.20%, and incremental allocation returns 4.59%. Incremental allocation reduces order events from 8,386 to 371 relative to equal resizing, but remains below the index. Allocation

Execution	Allocation	Return	Excess over index pp	Order events	Maximum drawdown
mid	independent	5.02%	−0.68	173	−10.84%
mid	equal	4.35%	−1.35	8386	−10.75%
mid	incremental	4.69%	−1.00	371	−10.15%
bid/ask	independent	4.90%	−0.80	173	−10.85%
bid/ask	equal	4.20%	−1.49	8386	−10.76%
bid/ask	incremental	4.59%	−1.11	371	−10.16%

Table E.1: Matched capital-allocation experiment using a 24-hour mean at 16:15, buy −0.20 and sell −0.90. Independent accounts total £10,000; shared portfolios start with £10,000. Events count buys and sells, including resizing, rather than decision dates.



Figure E.1: Daily returns under independent, equal-resizing and incremental capital rules, holding the 24-hour 16:15 signal and −0.20/−0.90 boundaries fixed, under midpoint (upper panel) and bid/ask (lower panel) execution. Each panel includes the observed FTSE 100 path.

affects exposure and concentration as well as spread costs; fewer trades do not guarantee the highest return. Order-event counts are also an imperfect cost measure: equal resizing uses 8,386 orders yet loses only about 0.15 percentage points more under bid/ask than at midpoint, because most resizing trades are small.

Shorter intervals react sooner but trade more; at midpoint this activity is free, whereas under bid/ask execution turnover creates a hurdle the signal must overcome.